Fe^{3+} is a form in which it can be bound to transport protein,

[transferrin] → for transport to body cells.

In a state of Cu deficiency, the transport of Fe for heme synthesis → decreases & iron can accumulate in tissues, =? where it can lead to long term organ damage.

[Zinc] The trace material/mineral functions as a co enzyme that facilitates the synthesis of the heme portion of hemoglobin.

- [B - vitamins] B vitamins → folate & B_{12} = function as co-enzymes that facilitate DNA synthesis critical for cells & for RBCs

[RBCs live up to 120 days]

worn out cells are removed by a type of myeloid phagocytic cell called → MACROPHAGE located primarily within the bone-marrow, liver, & spleen. The components of the degraded RBC → hemoglobin

- Globin → the protein portion of hemoglobin is broken down into amino acids → sent back to bone marrow → to be used to create new RBCs